

From the Hot Big Bang to Inflation, Reheating, and Model-Dependent Multiverse Scenarios

What is it? Why does it matter? How does it work? What do we actually know?



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Web study guide: <https://biswajit1999.github.io/Multiverse/learn/>

Research site: <https://biswajit1999.github.io/Multiverse/>

GitHub: <https://github.com/Biswajit1999/Multiverse>

How to read this monograph

This report is deliberately structured as a continuation: every chapter first gives one compact study page, followed by a full-page visual that reinforces the same topic. The study page always answers the same five questions: **What is it? Why does it matter? How does it work? What do we actually know? What remains open?** The order moves from the most observationally established material toward increasingly model-dependent physics. The hot Big Bang and the CMB come first. Inflation and reheating provide the project's working bridge into that hot state.

SCIENTIFIC STATUS RULE

A mathematical possibility is not the same thing as an observation. Direct measurements, calculations, model-dependent extrapolations and open questions are kept conceptually separate.

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The hot Big Bang: what it explains — and what it does not

WHAT IS IT?

The hot Big Bang is the successful description of an early universe that was hot, dense, expanding and close to homogeneous. It explains cooling, primordial nuclei, the cosmic microwave background and the later growth of structure. It is not best thought of as an explosion into pre-existing space.

WHY DOES IT MATTER?

The phrase Big Bang is often asked to do more than the model actually does. Standard hot-Big-Bang cosmology begins with an already hot expanding state. Asking what physical process produced that state is therefore a separate, earlier-layer question.

HOW DOES IT WORK?

For a homogeneous and isotropic universe, the scale factor tracks expansion. Einstein's equations reduce to the Friedmann equations. Matter, radiation, curvature and vacuum energy each alter the expansion history in different ways.

WHAT DO WE ACTUALLY KNOW?

The CMB blackbody spectrum, primordial light-element abundances, galaxy redshifts and large-scale structure strongly support a hot expanding history. Planck tightly constrains the standard cosmological model [12]. These observations establish the hot phase, not a unique prehistory.

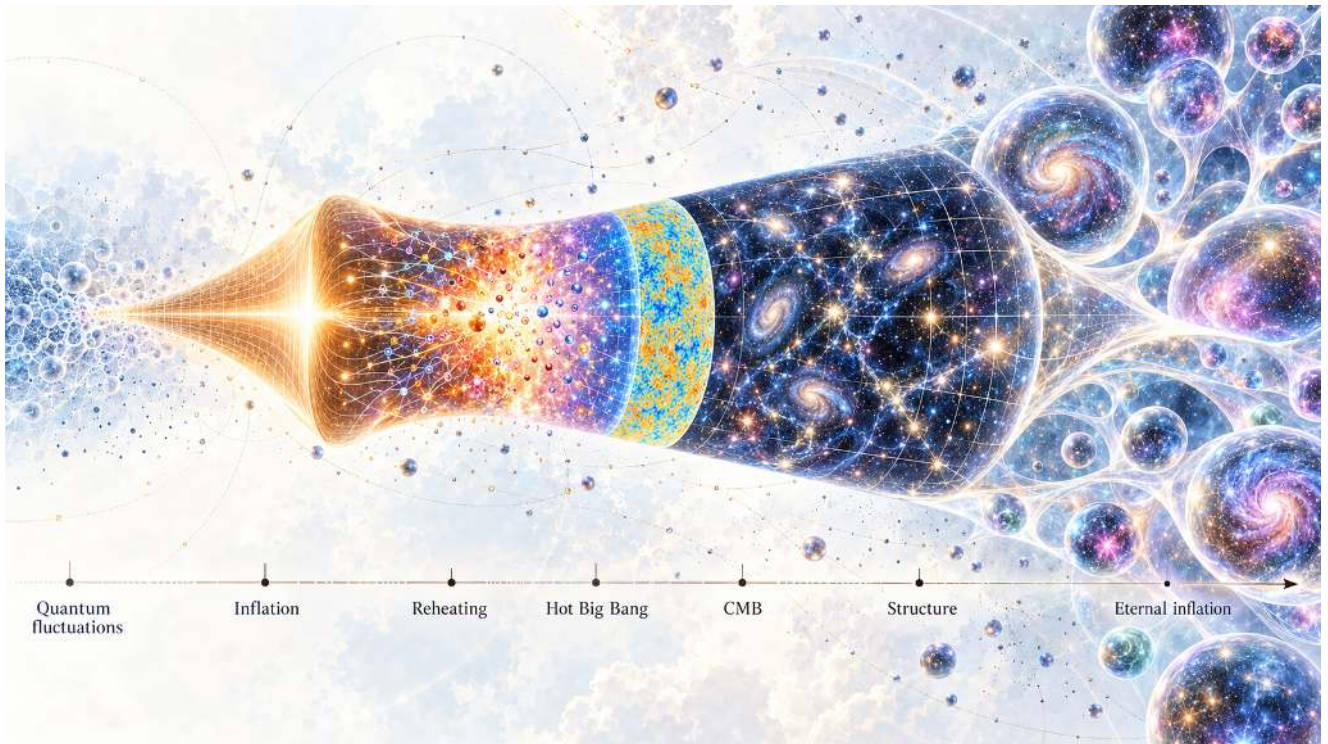
KEY EQUATION / DIAGNOSTIC

$$H^2 = \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{kc^2}{a^2} + \frac{\Lambda c^2}{3}$$

OPEN QUESTION

What physical mechanism generated the hot initial state, and where does the classical description stop being reliable?

VISUAL CONTINUATION



The hot Big Bang: what it explains — and what it does not

Visual continuation of Chapter 1. Conceptual research illustration unless explicitly identified as measured data.

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What could have come before the hot Big Bang?

WHAT IS IT?

The word *before* is subtle in cosmology. It can mean an earlier classical phase, a quantum boundary condition, a contracting universe, an inflating state, or an extrapolation beyond the domain in which present equations can be trusted.

WHY DOES IT MATTER?

Different models answer different versions of the question. A nonsingular bounce replaces a singular transition. Inflation supplies an earlier accelerated phase but does not automatically make the history past eternal. Quantum cosmology can replace an ordinary earlier time with a boundary condition on a wave function.

HOW DOES IT WORK?

A useful comparison asks: what are the degrees of freedom, what equations govern them, what state is assumed, does the model reach a hot radiation-dominated era, what perturbations survive, and what observations could distinguish it?

WHAT DO WE ACTUALLY KNOW?

Hawking–Penrose singularity theorems establish geodesic incompleteness under specified assumptions [6]. Borde–Guth–Vilenkin gives a different incompleteness result for sufficiently expanding spacetimes [7]. Neither result identifies the physical replacement for the boundary.

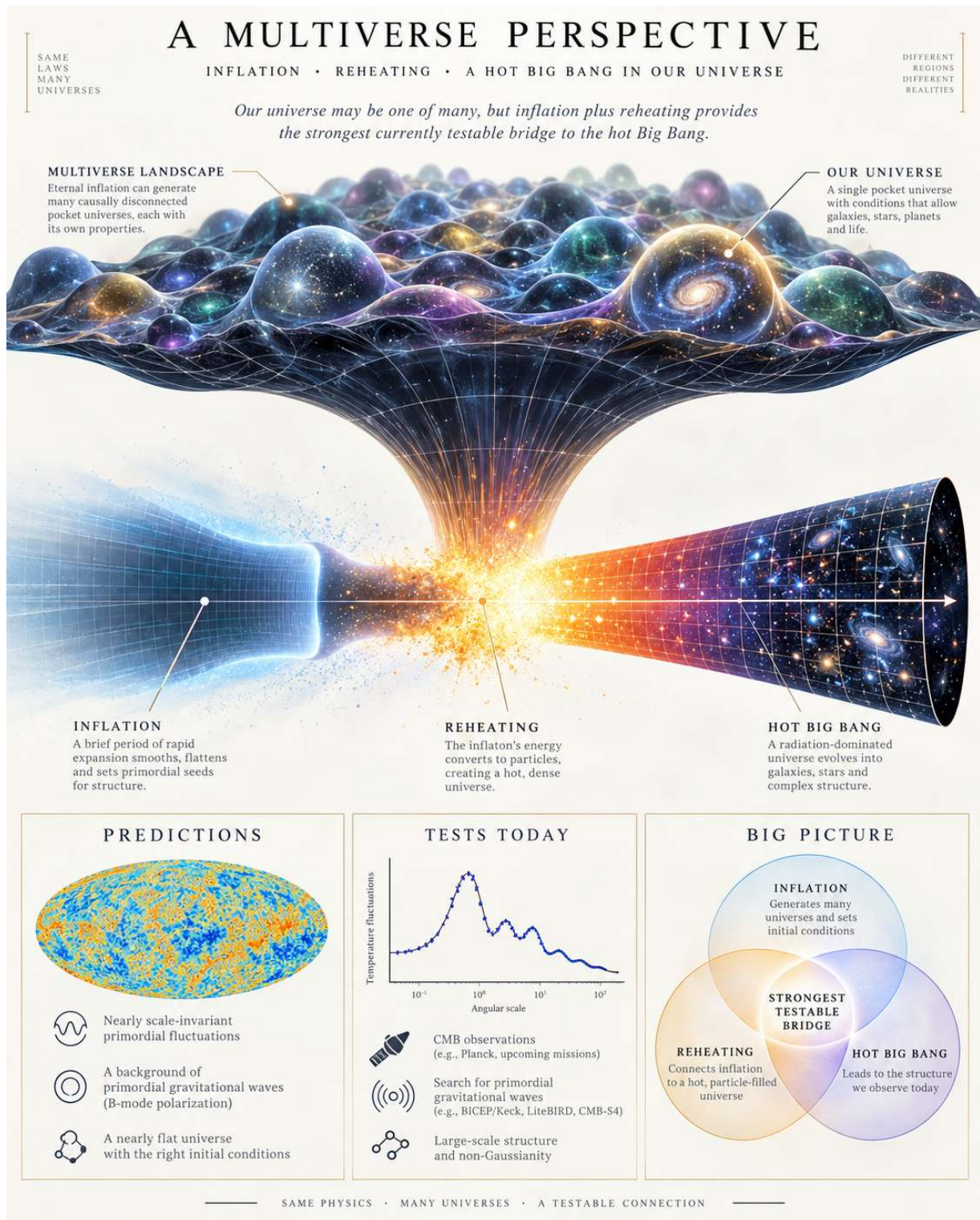
KEY EQUATION / DIAGNOSTIC

$$H_{\text{av}} > 0 \implies \text{past geodesic incompleteness in the BGV sense}$$

OPEN QUESTION

Is the earliest meaningful description classical, quantum, emergent, cyclic, bouncing, or something not yet formulated?

VISUAL CONTINUATION



What could have come before the hot Big Bang?

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Inflation: accelerated expansion before the hot phase

WHAT IS IT?

Inflation is a period in which the scale factor accelerates. In simple models a scalar field evolves slowly on a potential. When potential energy dominates over kinetic energy, the effective pressure becomes sufficiently negative to drive accelerated expansion.

WHY DOES IT MATTER?

Inflation was introduced to address the horizon and flatness problems and to dilute unwanted relics. Its deeper importance is that it also supplies a calculable mechanism for generating primordial fluctuations.

HOW DOES IT WORK?

The inflaton obeys a damped field equation in an expanding background. If the potential is sufficiently flat, Hubble friction keeps the field rolling slowly. Different potentials generate different slow-roll trajectories and therefore different observable predictions.

WHAT DO WE ACTUALLY KNOW?

Inflation is not directly photographed. It is tested through the primordial perturbation spectrum it predicts. Planck supports a nearly scale-invariant adiabatic scalar spectrum and constrains broad inflationary model classes [13].

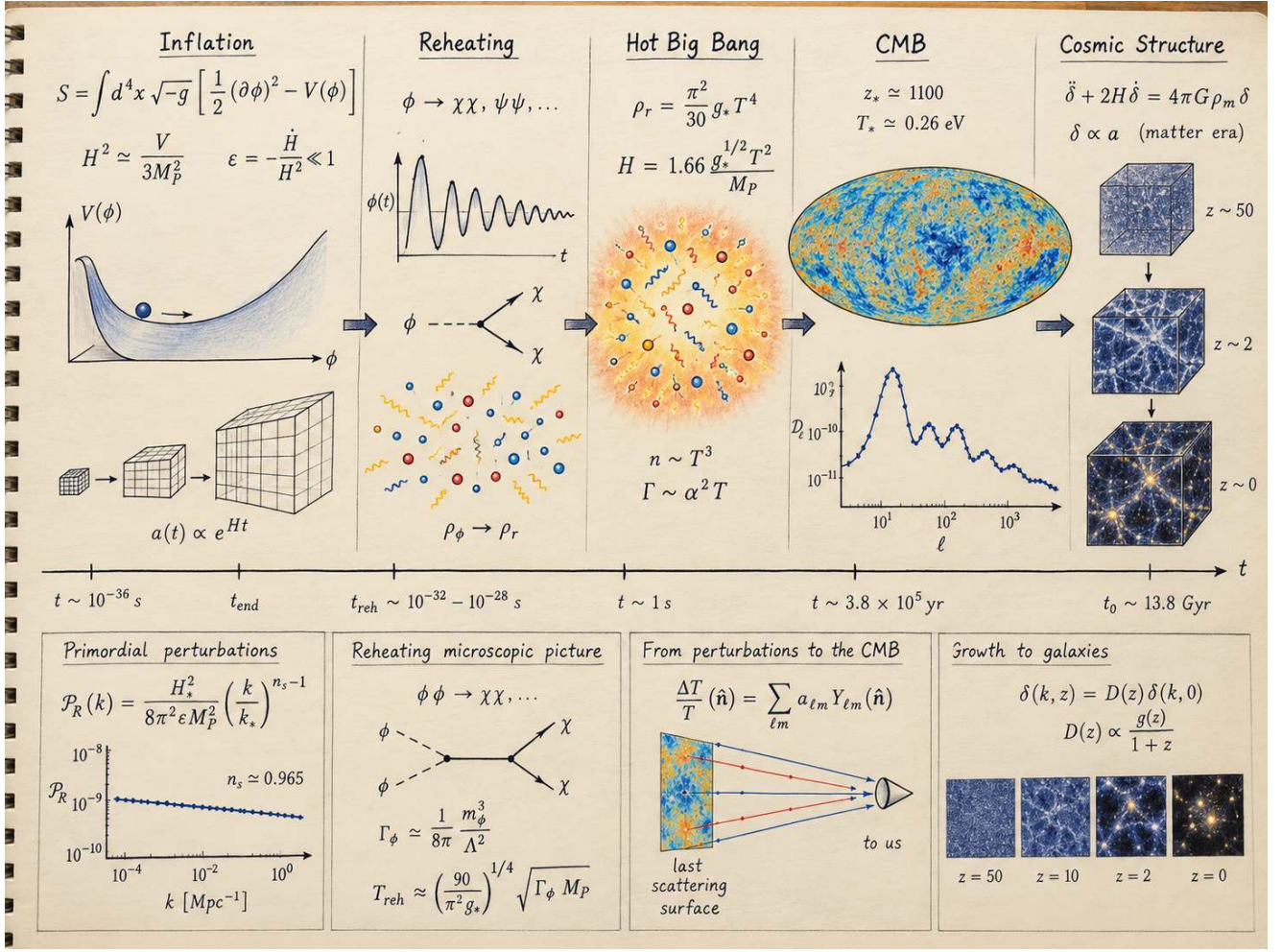
KEY EQUATION / DIAGNOSTIC

$$\ddot{\phi} + 3H\dot{\phi} + V_{,\phi} = 0, \quad \epsilon_V = \frac{M_{\text{Pl}}^2}{2} \left(\frac{V_{,\phi}}{V} \right)^2$$

OPEN QUESTION

What selected the inflating state and potential, and was inflation itself preceded by another phase?

VISUAL CONTINUATION



Inflation: accelerated expansion before the hot phase

Visual continuation of Chapter 3. Conceptual research illustration unless explicitly identified as measured data.

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Quantum fluctuations: how tiny seeds become cosmic structure

WHAT IS IT?

Inflation does more than alter the background expansion. Quantum fluctuations of the inflaton and spacetime are stretched to wavelengths much larger than the Hubble radius. After inflation they re-enter and become initial conditions for density and gravitational-potential perturbations.

WHY DOES IT MATTER?

Without a mechanism for generating initial inhomogeneity, a perfectly smooth universe would remain smooth. Galaxies, clusters and the cosmic web require seeds. Inflation provides a quantum origin for those seeds and links microscopic physics to astronomical structure.

HOW DOES IT WORK?

The gauge-invariant Mukhanov–Sasaki variable obeys a mode equation in conformal time. Vacuum fluctuations evolve through horizon crossing and freeze into an almost classical stochastic field whose power spectrum seeds later structure.

WHAT DO WE ACTUALLY KNOW?

The observed scalar spectrum is close to, but not exactly, scale invariant. Its phase coherence and approximately adiabatic nature fit the inflationary mechanism well. The formalism was developed in foundational work by Mukhanov and Sasaki [1, 2].

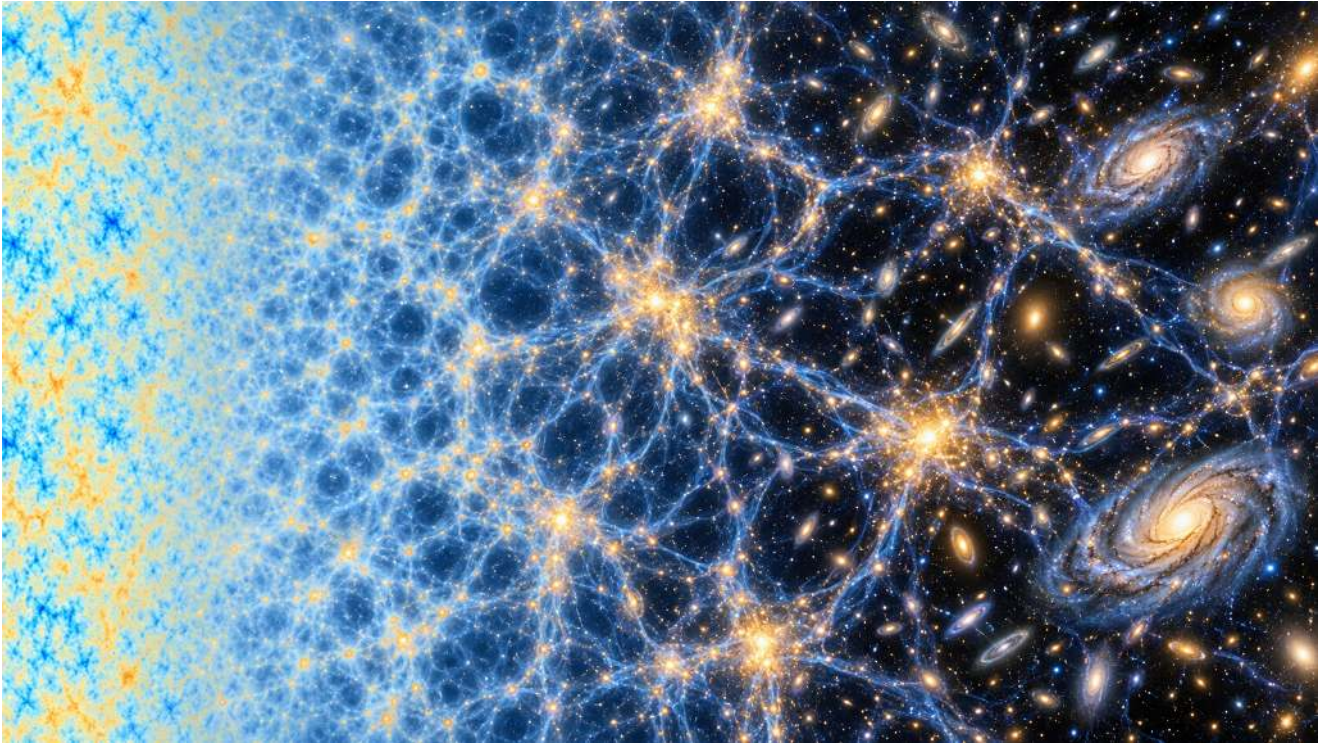
KEY EQUATION / DIAGNOSTIC

$$v_k'' + \left(k^2 - \frac{z''}{z} \right) v_k = 0, \quad \mathcal{P}_{\mathcal{R}} \simeq \frac{H_*^2}{8\pi^2 M_{\text{Pl}}^2 \epsilon_*}$$

OPEN QUESTION

Can future data reveal primordial non-Gaussianity, features, running or tensors that uniquely distinguish one mechanism from competitors?

VISUAL CONTINUATION



Quantum fluctuations: how tiny seeds become cosmic structure

Visual continuation of Chapter 4. Conceptual research illustration unless explicitly identified as measured data.

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Reheating: how inflation becomes a hot universe

WHAT IS IT?

Reheating is the transfer of energy from the inflationary sector into particles and radiation. Inflation leaves a universe dominated by the field driving accelerated expansion. The hot Big Bang requires a later state filled with ordinary excitations that can thermalize.

WHY DOES IT MATTER?

This step matters because the mapping between inflationary model parameters and observables depends on the post-inflationary expansion history. The reheating equation of state and reheating temperature therefore affect the inferred number of e-folds.

HOW DOES IT WORK?

A simple perturbative model treats the inflaton energy density as decaying at a rate Γ while radiation gains the same energy. In other models, coherent oscillations can drive explosive particle production through parametric resonance, known as preheating.

WHAT DO WE ACTUALLY KNOW?

Reheating has not been directly observed as a distinct event. Its consequences are constrained indirectly by successful nucleosynthesis and by consistency between inflationary predictions and later cosmology. Parametric preheating was developed by Kofman, Linde and Starobinsky [3].

KEY EQUATION / DIAGNOSTIC

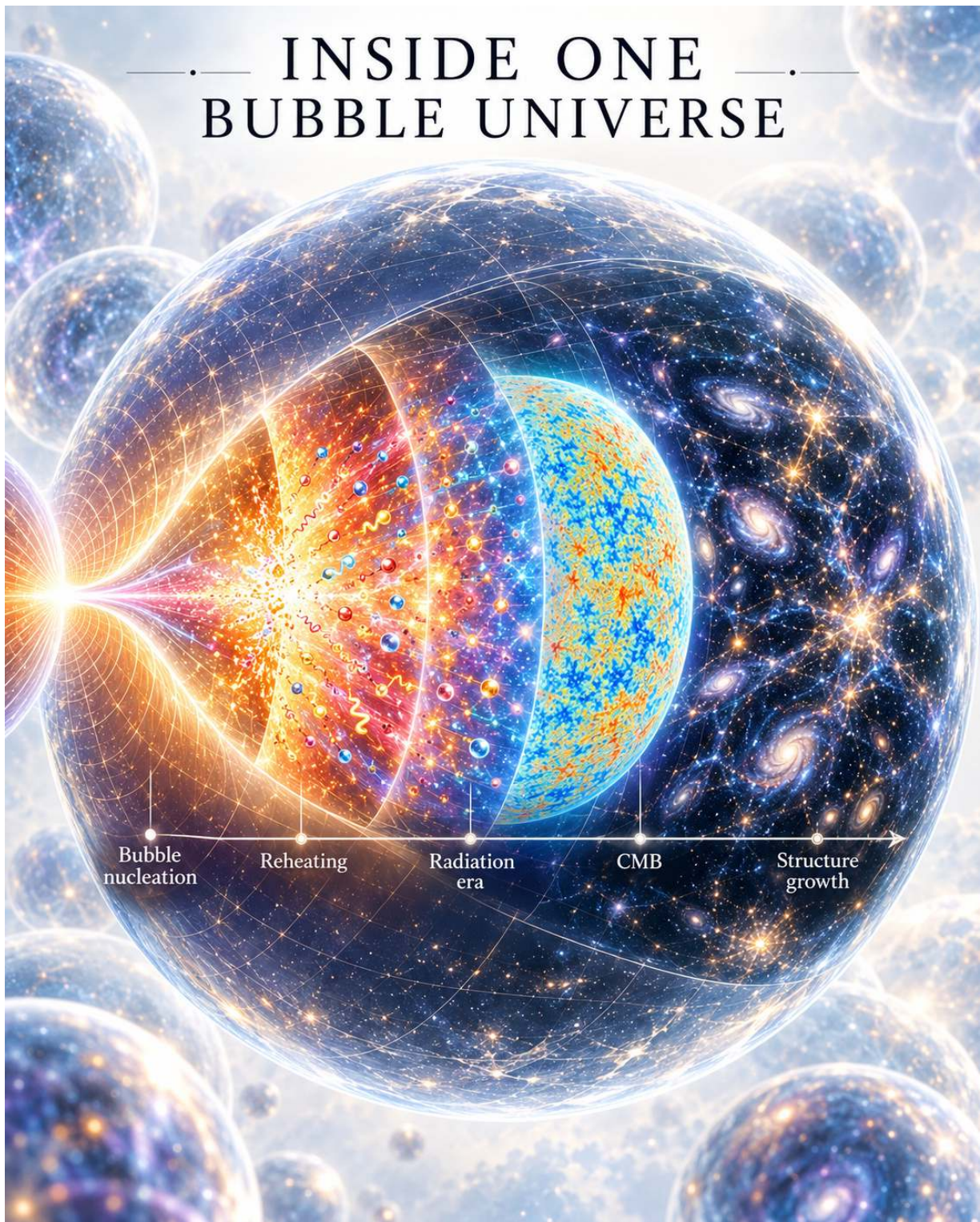
$$\dot{\rho}_\phi + 3H(1 + w_\phi)\rho_\phi = -\Gamma_\phi\rho_\phi, \quad \dot{\rho}_R + 4H\rho_R = +\Gamma_\phi\rho_\phi$$

OPEN QUESTION

Which fields received the inflaton's energy, what was the reheating temperature, and can any relic uniquely reveal that process?

VISUAL CONTINUATION

— . — INSIDE ONE — . — BUBBLE UNIVERSE



Reheating: how inflation becomes a hot universe

Visual continuation of Chapter 5. Conceptual research illustration unless explicitly identified as measured data.

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The CMB: the precision bridge from early theory to data

WHAT IS IT?

The cosmic microwave background is relic radiation released when the universe became transparent roughly 380,000 years after the hot Big Bang. Its temperature and polarization maps preserve statistical information about primordial perturbations and the photon–baryon plasma.

WHY DOES IT MATTER?

The CMB is powerful because the early plasma is governed by calculable linear physics over a wide range of scales. Peak positions, heights, damping and polarization respond to matter content, geometry, ionization history and the primordial spectrum.

HOW DOES IT WORK?

The sky is decomposed into spherical harmonics and summarized by angular power spectra. Temperature, E-mode polarization, their cross-spectrum, lensing and B modes provide complementary information.

WHAT DO WE ACTUALLY KNOW?

Planck, ACT, SPT and other experiments measure the CMB over overlapping scale ranges. NASA LAMBDA distributes public plotting data and spectra [15]. These measurements strongly support the standard cosmological framework while tightening early-universe parameter space.

KEY EQUATION / DIAGNOSTIC

$$D_\ell \equiv \frac{\ell(\ell+1)}{2\pi} C_\ell$$

OPEN QUESTION

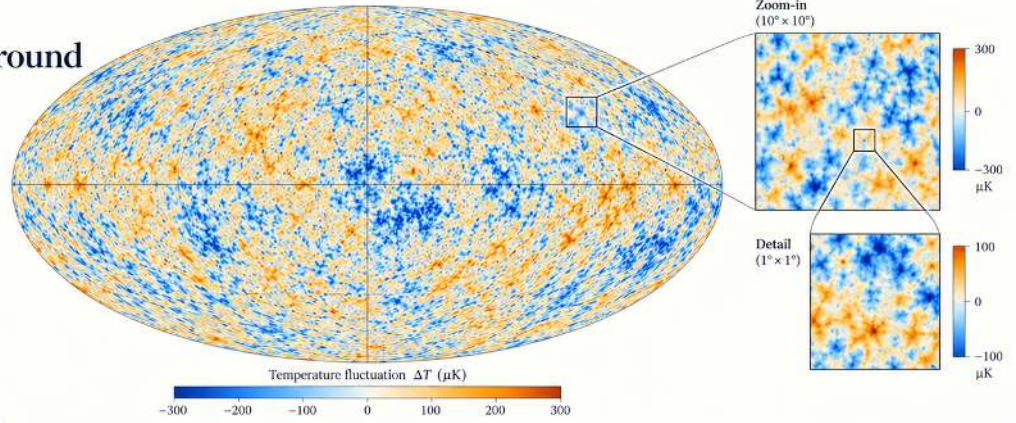
Will improved polarization, lensing and foreground control expose a primordial tensor signal or new departures from the minimal spectrum?

VISUAL CONTINUATION

The Cosmic Microwave Background

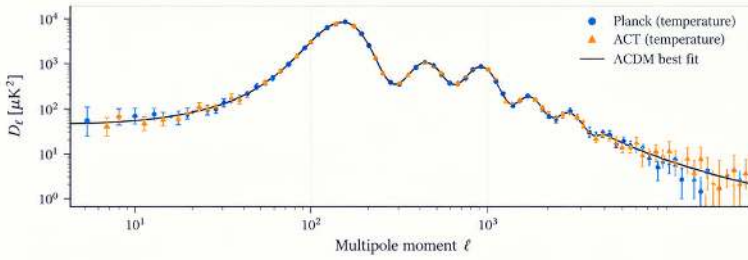
A fossil snapshot of the Universe $\approx 380,000$ years after the Big Bang, revealing tiny temperature fluctuations that seeded cosmic structure.

- $T = 2.725$ K
Mean temperature today
- $\Delta T \approx 10^{-5}$ K
Tiny fluctuations in temperature
- $\ell \approx 1 - 2500$
Angular scales probed by modern CMB experiments

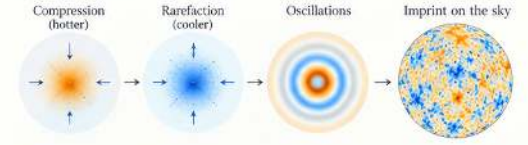


CMB Angular Power Spectrum

Acoustic peaks from sound waves in the early Universe

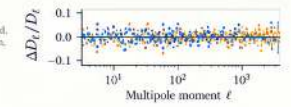


Acoustic oscillations in the photon-baryon plasma



A remarkable agreement

Independent measurements, from space and the ground, trace the same acoustic pattern with exquisite precision, confirming the standard cosmological model.



The CMB: the precision bridge from early theory to data

Visual continuation of Chapter 6. Conceptual research illustration unless explicitly identified as measured data.

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Inflation on trial: scalar tilt and tensor amplitude

WHAT IS IT?

The scalar spectral index describes how primordial scalar power changes with scale. The tensor-to-scalar ratio describes primordial tensor power relative to scalar power. Together they provide a compact way to compare inflationary models.

WHY DOES IT MATTER?

Different potentials that look equally plausible at the level of background expansion can predict very different tensor amplitudes or spectral tilts. Precision data can therefore eliminate models even when the broad idea of inflation remains viable.

HOW DOES IT WORK?

In first-order slow roll, the observables depend on derivatives of the potential evaluated when observable modes left the Hubble radius. Varying the number of e-folds traces a model trajectory in the observable plane.

WHAT DO WE ACTUALLY KNOW?

Planck 2018 found a scalar tilt below exact scale invariance and strongly constrained many models [13]. BICEP/Keck data place stringent limits on primordial tensor power [14].

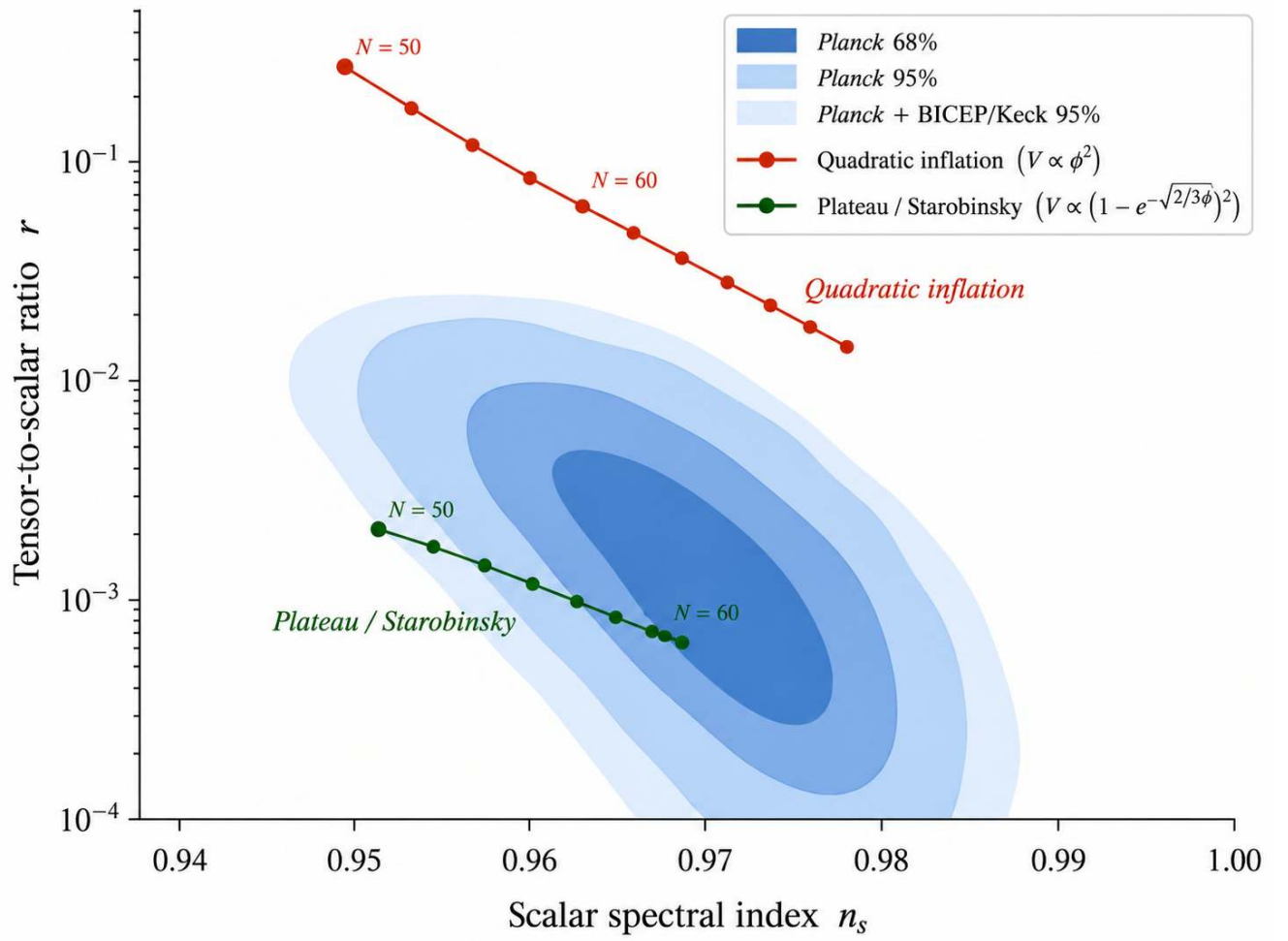
KEY EQUATION / DIAGNOSTIC

$$n_s \simeq 1 - 6\epsilon_V + 2\eta_V, \quad r \simeq 16\epsilon_V$$

OPEN QUESTION

How far can future polarization and gravitational-wave experiments reduce the allowed inflationary model space?

VISUAL CONTINUATION



Inflation on trial: scalar tilt and tensor amplitude

Visual continuation of Chapter 7. Conceptual research illustration unless explicitly identified as measured data.

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False-vacuum decay: the field theory behind cosmic bubbles

WHAT IS IT?

A false vacuum is a metastable local minimum of a field potential. It can be classically long lived yet decay quantum mechanically to a lower-energy state. In cosmology, the new phase appears as a nucleated region whose subsequent geometry and field evolution must be calculated.

WHY DOES IT MATTER?

The phrase bubble universe often appears visually before the mechanism is explained. The correct order is the reverse: define the potential, identify the vacua, solve the tunnelling problem, calculate the decay action, and only then ask what the interior region becomes.

HOW DOES IT WORK?

Coleman–De Luccia vacuum decay is described by an $O(4)$ -symmetric Euclidean bounce including gravity. The decay rate depends exponentially on the Euclidean action difference, making the transition rate extremely sensitive to the potential.

WHAT DO WE ACTUALLY KNOW?

Vacuum decay is a well-defined semiclassical calculation in quantum field theory. Coleman and De Luccia incorporated gravity into the problem [4]. A cosmological transition of the type required by a specific multiverse model has not been directly observed.

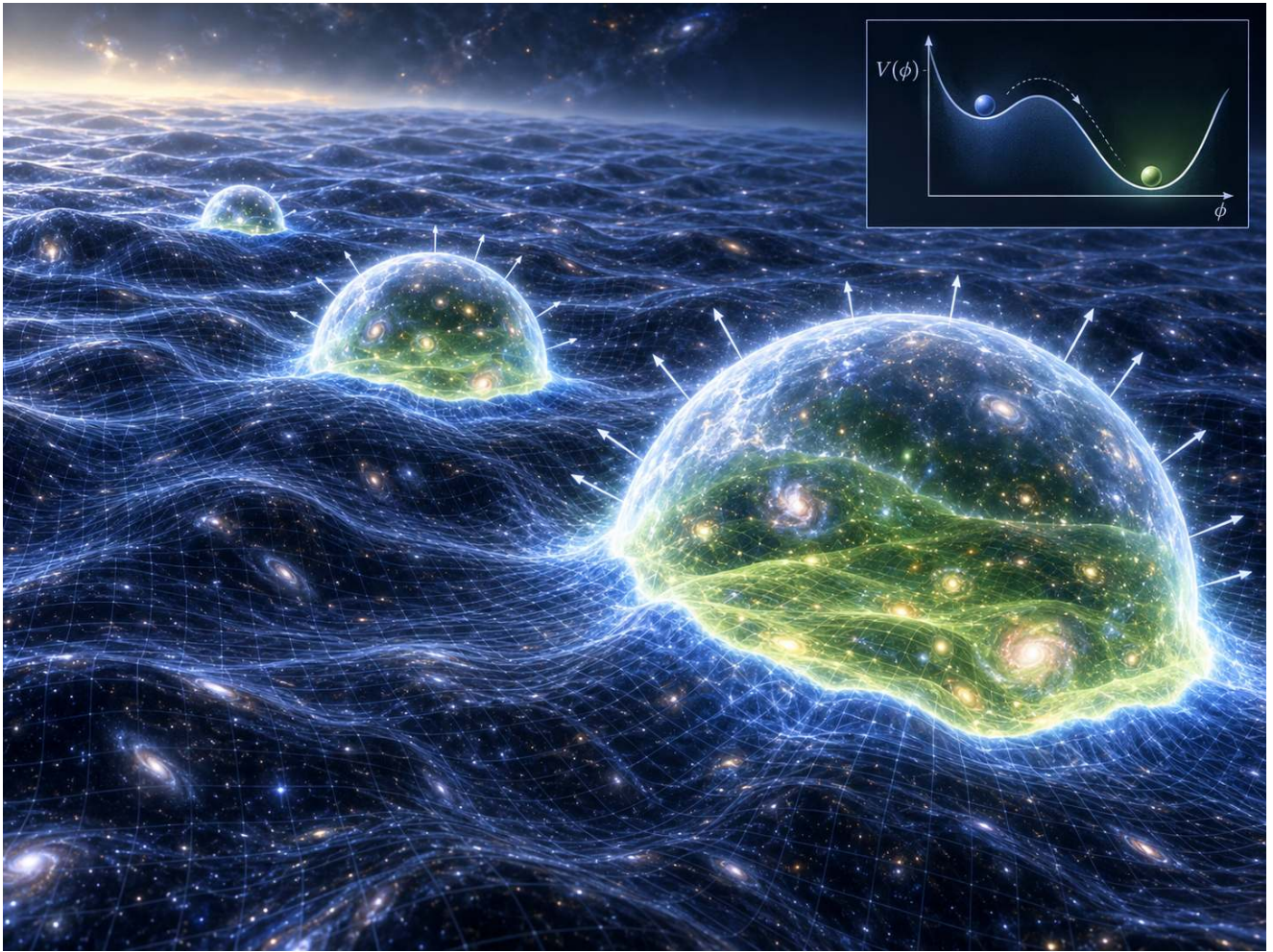
KEY EQUATION / DIAGNOSTIC

$$\frac{\Gamma}{V} \sim A \exp\left(-\frac{B}{\hbar}\right), \quad B = S_E[\text{bounce}] - S_E[\text{false}]$$

OPEN QUESTION

What high-energy potential, if any, describes our vacuum history, and could a transition leave an observable relic?

VISUAL CONTINUATION



False-vacuum decay: the field theory behind cosmic bubbles

Visual continuation of Chapter 8. Conceptual research illustration unless explicitly identified as measured data.

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Vacuum landscapes and transition networks

WHAT IS IT?

A vacuum landscape is a theory-space picture in which many metastable field configurations correspond to different effective low-energy physics. Transitions between them may occur through tunnelling or de Sitter fluctuations.

WHY DOES IT MATTER?

If a fundamental theory admits many vacua, an eternally inflating spacetime could populate more than one effective vacuum. The problem then becomes dynamical: which vacua are accessible, how often, and with what low-energy properties?

HOW DOES IT WORK?

A transition network requires more than a list of minima. One needs barrier shapes, decay rates, gravitational corrections, expansion rates and a prescription for comparing the abundance of regions.

WHAT DO WE ACTUALLY KNOW?

Landscape reasoning is theoretical. No astronomical survey maps vacuum labels across a multiverse. Its scientific value comes from whether a proposed high-energy framework genuinely yields such states and whether any resulting low-energy predictions can be constrained.

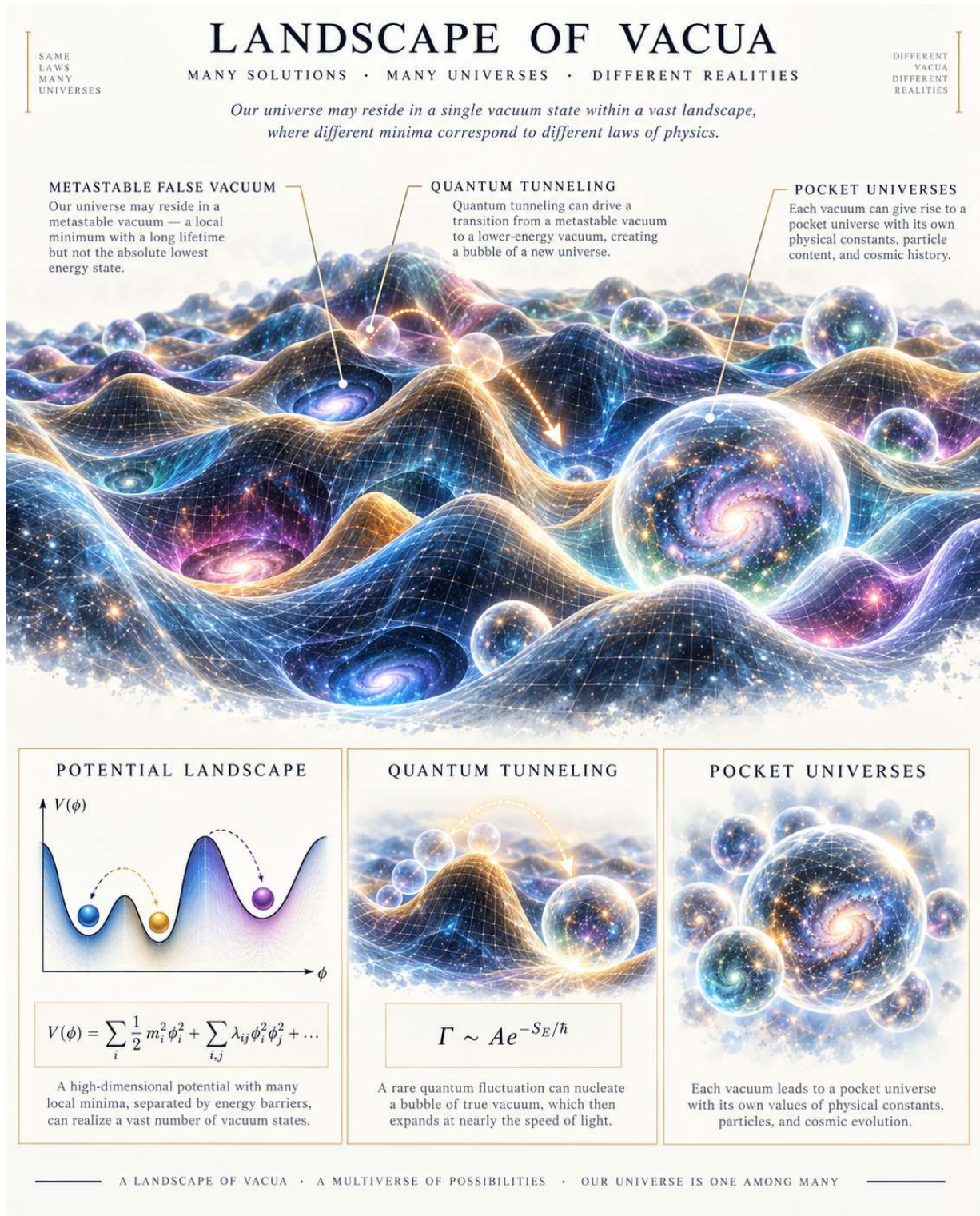
KEY EQUATION / DIAGNOSTIC

$$\Gamma_{i \rightarrow j}/V \sim A_{ij} \exp(-B_{ij}/\hbar)$$

OPEN QUESTION

Can a landscape produce robust probabilities without an arbitrary measure prescription, and can those probabilities ever be confronted with observation?

VISUAL CONTINUATION



Vacuum landscapes and transition networks

Visual continuation of Chapter 9. Conceptual research illustration unless explicitly identified as measured data.

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Eternal inflation: when inflation can become self-reproducing

WHAT IS IT?

Eternal inflation is a regime in which inflation ends locally but continues globally. Quantum fluctuations can push the inflating field upward in some Hubble patches while classical rolling drives other patches toward reheating. Rapid expansion allows the inflating volume to keep growing.

WHY DOES IT MATTER?

This is one route by which a multiverse can emerge as a consequence rather than a starting assumption. Ordinary inflation and eternal inflation are not synonymous; the self-reproduction condition must be checked for each model.

HOW DOES IT WORK?

For a light scalar during inflation, a characteristic quantum step is proportional to H . The classical drift depends on the potential slope. Where the quantum step becomes sufficiently important, stochastic evolution can dominate the local classical roll.

WHAT DO WE ACTUALLY KNOW?

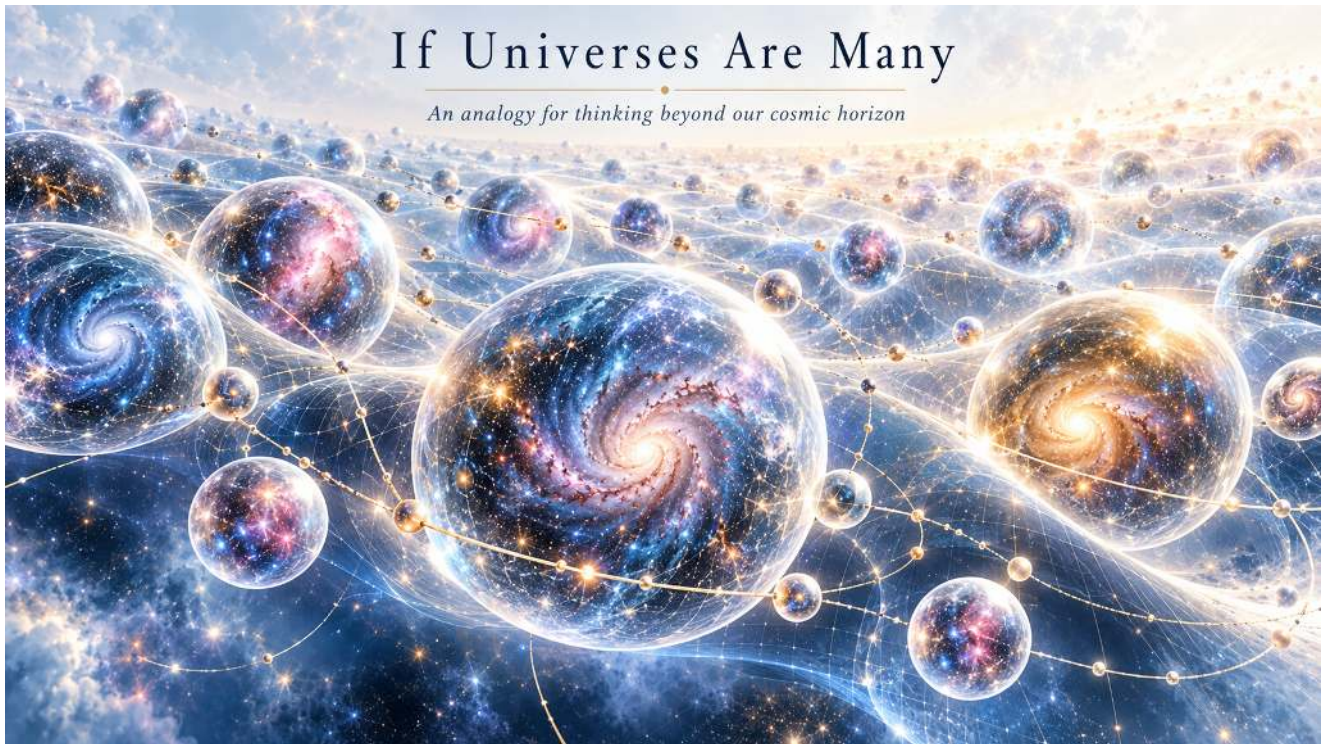
Self-reproduction is a theoretical property of specified potentials and stochastic dynamics [5]. No external pocket universe has been directly observed. Proposed signatures are model specific and must compete with ordinary statistical fluctuations and foreground explanations.

KEY EQUATION / DIAGNOSTIC

$$\delta\phi_q \simeq \frac{H}{2\pi}, \quad \mathcal{R}_{\text{EI}} \simeq \frac{3H^3}{2\pi|V_{,\phi}|}$$

OPEN QUESTION

If eternal inflation occurs, how should probabilities be defined in an ever-growing spacetime with an infinite number of regions?



Eternal inflation: when inflation can become self-reproducing

Visual continuation of Chapter 10. Conceptual research illustration unless explicitly identified as measured data.

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Cosmic horizons: where observation ends

WHAT IS IT?

A cosmological horizon is a causal boundary, not a material wall. The particle horizon limits how far signals could have travelled to us since an initial time. An event horizon, when present, limits what signals sent now will ever reach us in the future.

WHY DOES IT MATTER?

This distinction prevents a common mistake: equating beyond the observable universe with another universe. A region can be unseen while still belonging to the same connected FLRW spacetime.

HOW DOES IT WORK?

Conformal time measures the causal distance light can traverse in an expanding universe. Integrating one over the scale factor into the past gives the particle horizon; integrating into the future determines whether an event horizon exists.

WHAT DO WE ACTUALLY KNOW?

Horizons are calculated from the measured and modelled expansion history. The late-time accelerated expansion inferred in the standard cosmological model implies an event-horizon-like causal structure if that acceleration persists indefinitely.

KEY EQUATION / DIAGNOSTIC

$$\eta(t) = \int^t \frac{dt'}{a(t')}, \quad \chi_p(t) = c \int_{t_i}^t \frac{dt'}{a(t')}$$

OPEN QUESTION

Does the global spacetime extend far beyond our observable patch, and if so, are its distant regions physically identical to ours?

VISUAL CONTINUATION



Cosmic horizons: where observation ends

Visual continuation of Chapter 11. Conceptual research illustration unless explicitly identified as measured data.

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Nonsingular bounce cosmology

WHAT IS IT?

A bounce model replaces a singular classical beginning with an earlier contracting phase that reaches a minimum size or maximum density and then re-expands. The bounce can arise from modified gravity, quantum geometry, exotic matter or other high-density effects.

WHY DOES IT MATTER?

A successful bounce could provide a prehistory without requiring the classical scale factor to vanish. But avoiding a singular background is only the first step: the model must also explain homogeneity, anisotropy, perturbations and the transition to the observed hot universe.

HOW DOES IT WORK?

In effective loop quantum cosmology, a frequently used modified Friedmann equation adds a correction that forces the Hubble rate to vanish at a critical density, allowing a turnaround in the effective dynamics.

WHAT DO WE ACTUALLY KNOW?

No cosmological bounce has been directly observed. Effective loop quantum cosmology provides a controlled example of modified high-density dynamics [10]. Models are tested through predicted spectra, non-Gaussianity, gravitational waves or other relics.

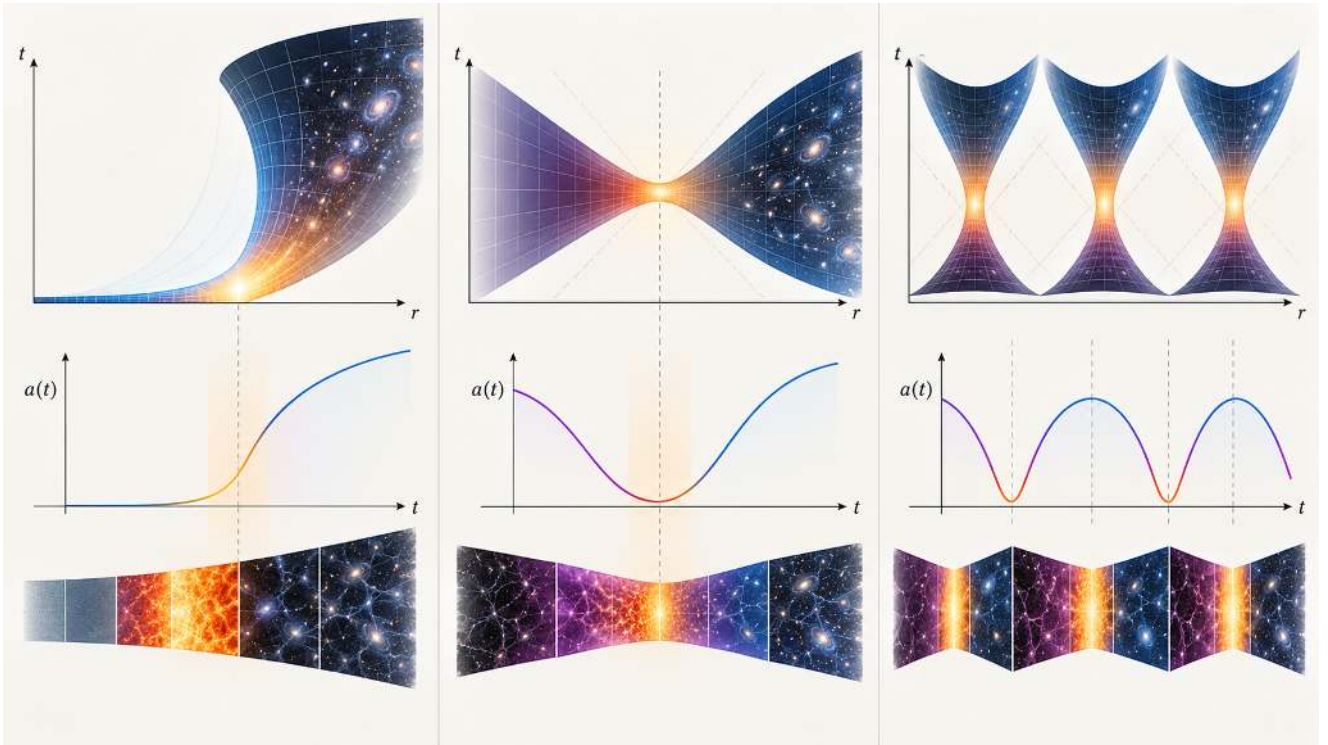
KEY EQUATION / DIAGNOSTIC

$$H^2 = \frac{8\pi G}{3} \rho \left(1 - \frac{\rho}{\rho_c} \right)$$

OPEN QUESTION

Can a complete bounce model make distinctive, robust predictions that outperform inflationary explanations of the same data?

VISUAL CONTINUATION



Nonsingular bounce cosmology

Visual continuation of Chapter 12. Conceptual research illustration unless explicitly identified as measured data.

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Cyclic and ekpyrotic cosmology

WHAT IS IT?

Cyclic cosmologies explore repeated periods of expansion and contraction. Ekpyrotic models use a slowly contracting phase with a very stiff effective equation of state, often as part of a broader cyclic construction.

WHY DOES IT MATTER?

A contracting phase can smooth the universe in a way mathematically different from inflation. In an ekpyrotic regime, anisotropies can be suppressed relative to the dominant background, offering an alternative mechanism for preparing simple initial conditions.

HOW DOES IT WORK?

The dynamics depend on scalar fields, potentials and a controlled transition through the bounce. Generating the observed nearly scale-invariant curvature perturbations generally requires additional structure, such as entropy perturbations later converted to curvature perturbations.

WHAT DO WE ACTUALLY KNOW?

Cyclic and ekpyrotic models remain active theoretical alternatives. A well-known cyclic construction was developed by Steinhardt and Turok [11]. Current observations do not uniquely select a cyclic origin.

KEY EQUATION / DIAGNOSTIC

$$w \gg 1 \quad \text{during ekpyrotic contraction}$$

OPEN QUESTION

Can a cyclic model control entropy, anisotropy and the bounce across repeated cycles while preserving predictive perturbations?

VISUAL CONTINUATION



Cyclic and ekpyrotic cosmology

Visual continuation of Chapter 13. Conceptual research illustration unless explicitly identified as measured data.

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Quantum cosmology: when the origin becomes a boundary condition

WHAT IS IT?

Quantum cosmology applies quantum principles to the universe as a whole. Instead of asking for a classical trajectory all the way to an initial singularity, one may describe possible geometries and fields by a wave function subject to a quantum gravitational constraint.

HOW DOES IT WORK?

In canonical quantum gravity the Hamiltonian constraint becomes a Wheeler–DeWitt equation. In minisuperspace, the problem is reduced to a few variables such as the scale factor. Different proposals impose different boundary conditions on the wave function.

WHY DOES IT MATTER?

Near Planckian densities, classical general relativity is expected to be incomplete. A quantum description might replace the classical singularity or redefine what counts as an initial condition. In such a framework, asking what happened earlier may not be the most fundamental question.

WHAT DO WE ACTUALLY KNOW?

The Hartle–Hawking no-boundary proposal [8] and Vilenkin’s tunnelling proposal [9] are mathematically distinct constructions. Neither is empirically established as the unique quantum state of the universe.

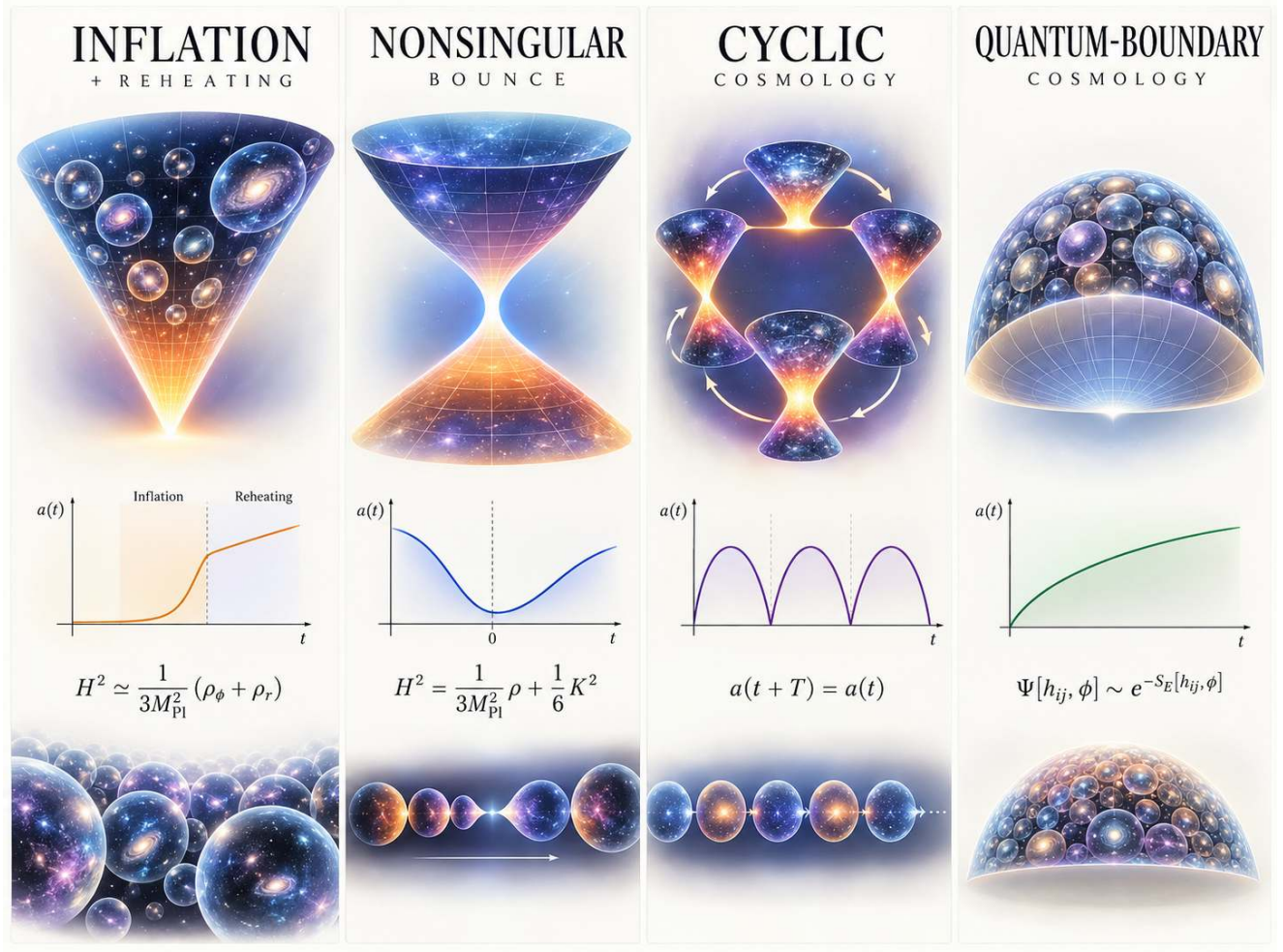
KEY EQUATION / DIAGNOSTIC

$$\hat{\mathcal{H}}\Psi = 0, \quad \left[-\frac{d^2}{da^2} + U(a) \right] \Psi(a) = 0$$

OPEN QUESTION

What is the correct quantum theory of gravity, what selects its boundary condition, and how do classical spacetime and probability emerge from it?

VISUAL CONTINUATION



Quantum cosmology: when the origin becomes a boundary condition

Visual continuation of Chapter 14. Conceptual research illustration unless explicitly identified as measured data.

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Could there be a higher-dimensional outside?

WHAT IS IT?

Standard cosmology describes spacetime intrinsically; it does not require the universe to sit inside an ordinary external room. Some theories, however, embed lower-dimensional spacetimes or branes in a higher-dimensional bulk. In those models, an outside can have a precise mathematical meaning.

HOW DOES IT WORK?

The useful distinction is between intrinsic geometry, measured by observers within spacetime, and extrinsic or embedding geometry, defined relative to a larger mathematical space. A brane-world model must specify how matter, gravity and extra dimensions interact.

WHY DOES IT MATTER?

The idea is attractive because human intuition likes to picture a universe as an object with an exterior. But intuition can outrun physics. An embedding space is not automatically a physically accessible location, and it does not imply observers living beyond our universe.

WHAT DO WE ACTUALLY KNOW?

Extra-dimensional models are constrained through laboratory, collider, gravitational and cosmological tests, but no observation establishes a literal external observer or viewpoint outside the whole universe. The next-page image is explicitly a conceptual thought experiment.

KEY EQUATION / DIAGNOSTIC

intrinsic geometry $\neq$ evidence for a literal external vantage point

OPEN QUESTION

If extra dimensions exist, what measurable signatures distinguish them from ordinary four-dimensional cosmology?

VISUAL CONTINUATION



Could there be a higher-dimensional outside?

Visual continuation of Chapter 15. Conceptual research illustration unless explicitly identified as measured data.

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Inside one bubble universe

WHAT IS IT?

A nucleated bubble or reheated pocket is not automatically a viable universe. Its interior must contain a cosmological solution that evolves from the transition into a state compatible with radiation domination, nucleosynthesis, recombination and structure formation.

WHY DOES IT MATTER?

This local history is what connects a global eternal-inflation picture to observers like us. Without a successful interior cosmology, a bubble is only a transition in field space. The interesting question is whether it produces a long-lived region with sensible low-energy physics.

HOW DOES IT WORK?

After nucleation, the bubble interior can possess an open FLRW geometry. Depending on the potential, it may undergo additional slow-roll inflation that dilutes curvature and prepares perturbations. Reheating then populates the interior with particles and radiation.

WHAT DO WE ACTUALLY KNOW?

This is a calculable feature of specific vacuum-decay models, not an observed map of our surroundings. Any claim that our universe is such a bubble must ultimately be tied to observables such as curvature, collision signatures or correlated primordial features.

KEY EQUATION / DIAGNOSTIC

$$ds^2 = -dt^2 + a^2(t) [d\chi^2 + \sinh^2 \chi d\Omega^2]$$

OPEN QUESTION

Could any relic of bubble nucleation survive inflation and later cosmic evolution strongly enough to be identified unambiguously?

VISUAL CONTINUATION



Inside one bubble universe

Visual continuation of Chapter 16. Conceptual research illustration unless explicitly identified as measured data.

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How could we test the earliest universe?

WHAT IS IT?

Early-universe cosmology is testable only through relics that survive into observable epochs. Different probes sample different scales, epochs and physical effects. Combining them is more powerful than asking one experiment to solve the origin problem alone.

WHY DOES IT MATTER?

Broad ideas such as inflation or bounce contain many models. A useful scientific programme identifies observables that vary across those models and designs measurements that can eliminate parameter space or discover a distinctive signature.

HOW DOES IT WORK?

CMB temperature and polarization constrain the primordial spectrum on large scales. Large-scale structure extends the lever arm. 21-cm cosmology could probe much smaller scales. Primordial gravitational waves, non-Gaussianity and stochastic backgrounds offer qualitatively different information.

WHAT DO WE ACTUALLY KNOW?

Modern datasets already eliminate many simple possibilities. The website's public-data observatory uses NASA/LAMBDA plotting tables for transparent visualization, while full parameter inference requires experiment likelihoods, covariance matrices and foreground models.

KEY EQUATION / DIAGNOSTIC

model $\rightarrow$ primordial perturbations $\rightarrow$ transfer physics $\rightarrow$ observable likelihood

OPEN QUESTION

Which future observation would force the current working synthesis to change?

VISUAL CONTINUATION



How could we test the earliest universe?

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The long future and the current research synthesis

WHAT IS IT?

Cosmology is a history in both directions. The same scale factor that governs the early universe also determines the far future. If accelerated expansion persists, more distant comoving regions become causally inaccessible while gravitationally bound structures remain local.

WHY DOES IT MATTER?

The far future clarifies what horizons actually mean and prevents a misleading picture of the universe as a static container. Causal accessibility changes with time. What we can observe is a property of spacetime history, not simply telescope power.

HOW DOES IT WORK?

An event horizon exists when the future conformal-time integral converges. In a universe asymptotically dominated by positive vacuum energy, accelerated expansion limits future communication across sufficiently large distances.

WHAT DO WE ACTUALLY KNOW?

Within this project, inflation followed by reheating remains the strongest implemented bridge from a pre-hot dynamical phase to the hot structured universe because it supplies both background evolution and calculable perturbations. This is a working synthesis, not proof that inflation is unique or that a multiverse exists.

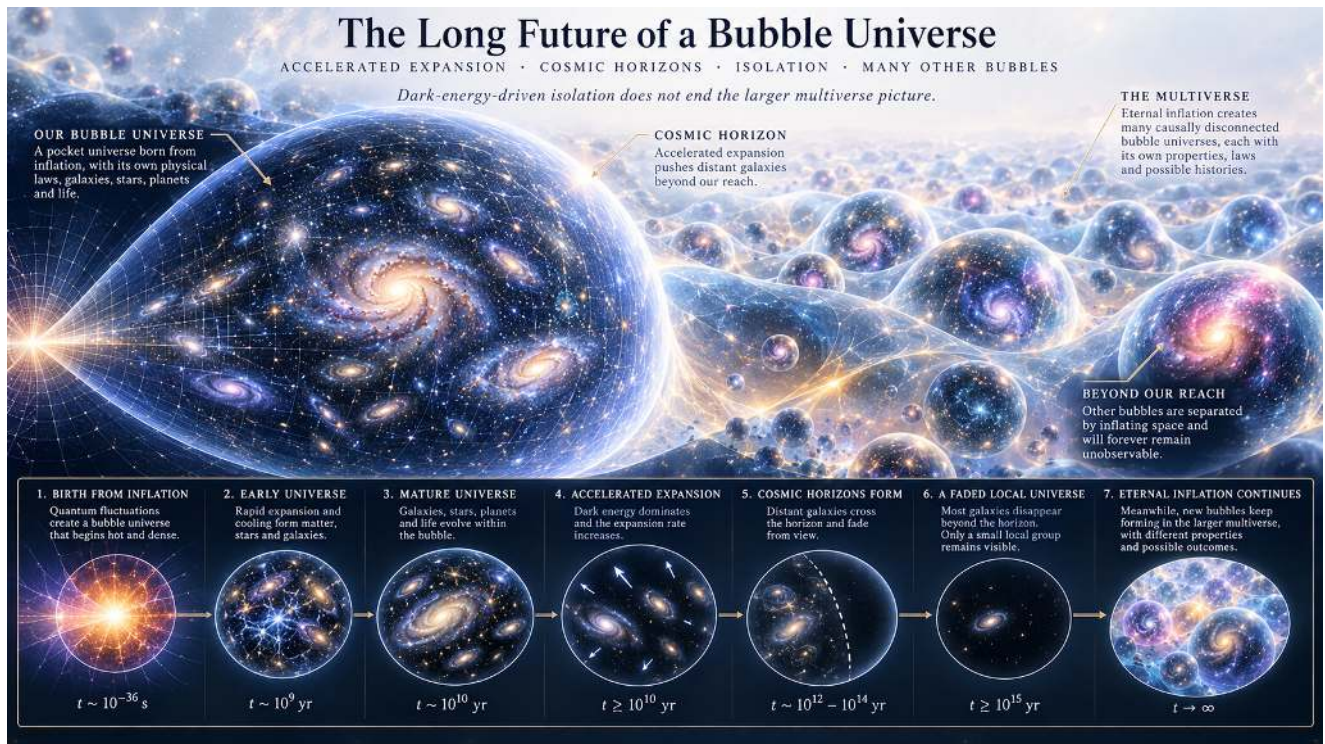
KEY EQUATION / DIAGNOSTIC

$$\chi_e(t) = c \int_t^\infty \frac{dt'}{a(t')}$$

OPEN QUESTION

What selected the inflating state, whether a deeper prehistory exists, and whether a global completion contains other universes remain open frontier questions.

VISUAL CONTINUATION

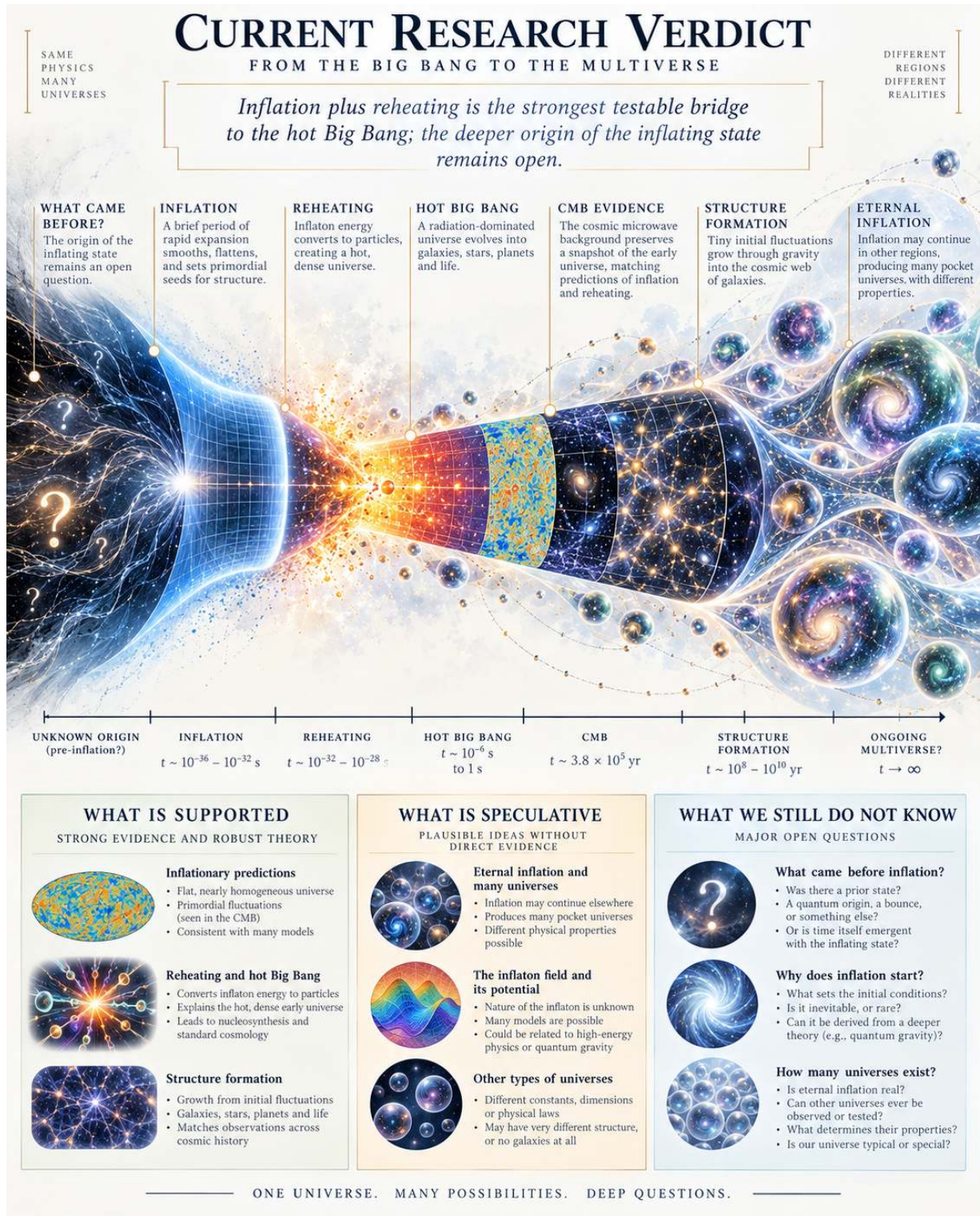


The long future and the current research synthesis

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VISUAL CONTINUATION



Current research verdict

Inflation followed by reheating is the strongest testable bridge implemented in this project. The deeper origin of the inflating state and the physical status of multiverse extensions remain open.

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Research practice, reproducibility and limits

The repository keeps numerical experiments, public data visualizations, model notes, tests and visual explanations together so that each claim can be traced to an equation, a data source, a calculation or an explicitly labelled conceptual interpretation. The browser-based CMB observatory uses public NASA/LAMBDA plotting tables for transparent visualization. Those tables are not substitutes for the full likelihood products used by the collaborations. Publication-grade cosmological inference requires covariance matrices, calibration and foreground nuisance parameters, masks, transfer functions and experiment-specific likelihood software. The conceptual figures are explanatory artwork. They are not telescope images of inflation, other universes, vacuum landscapes or higher-dimensional observers.

Copyright and citation

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Jana, B. (2026). *Multiverse Cosmology Lab: From the Hot Big Bang to Inflation, Reheating, and Model-Dependent Multiverse Scenarios*. <https://biswajit1999.github.io/Multiverse/>

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